

Panel MS-AR(1) with quarterly dummies, 1995Q1 onward

Real GDP per worker (period-average USD)

1 Model

Joint panel Markov-switching AR(1) around a linear trend with common quarterly dummies (Q1 omitted):

$$y_{it} = a_i + g t + d_2 Q2_t + d_3 Q3_t + d_4 Q4_t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}). \quad (3)$$

The sample is restricted to calendar time from 1995Q1. Specification 1 sets $a_i \equiv a$. Specification 2 lets $a_i \sim \mathcal{N}(\alpha, \omega^2)$, integrating each country by 11-point Gauss–Hermite. The slope g and the quarterly dummies are common. $|\rho| < 0.99$. Median μ pinned at 0. Cycles use posterior-mean a_i in the RE specification. Standard errors are delta-method from a numerical Hessian.

2 Common intercept, linear trend, and quarterly dummies

2026-09-14 panel from 1995Q1: log real GDP per worker. 90 countries, 7975 observations, 1995-Q1–2026-Q2. Fitted countries: 90. Observations: 7975. Log-likelihood: 12436.63. Ergodic mean of the cycle $E[z] = 0.4724$ (0.0941). Common intercept $a = -5.4017$ (0.0771). Common slope $g = -0.0018$ (0.0016). Seasonal dummies (Q1 omitted): $d_{Q2} = 0.0177$ (0.0009), $d_{Q3} = 0.0128$ (0.0010), $d_{Q4} = 0.0432$ (0.0009).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-1.1440 (0.3007)	0.0000 —	1.5060 (0.0708)
σ	0.1693 (0.0050)	0.0654 (0.0013)	0.0274 (0.0005)
ρ	0.9900	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.9825 (0.0060)	0.0149 (0.0050)	0.0026 (0.0032)
Π_1	0.0039 (0.0013)	0.9808 (0.0029)	0.0153 (0.0026)
Π_2	0.0015 (0.0012)	0.0158 (0.0026)	0.9827 (0.0023)
π	0.1357 (0.0289)	0.4474 (0.0354)	0.4169 (0.0363)

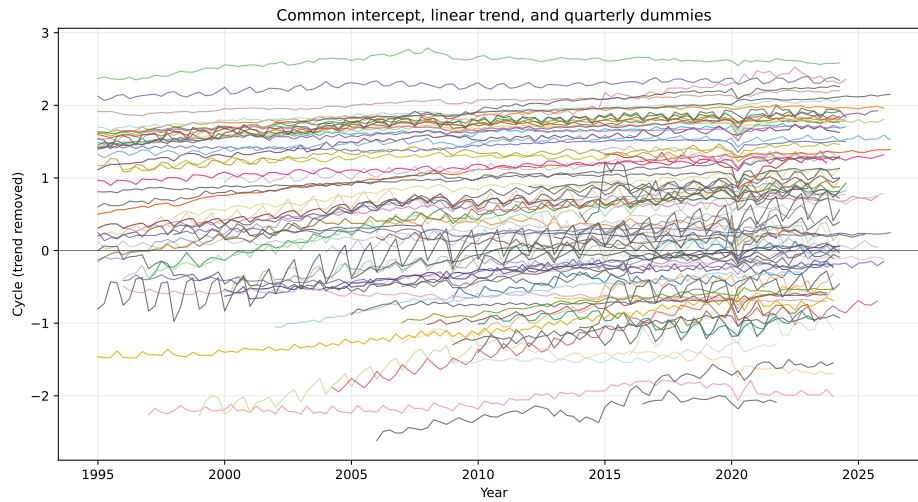


Figure 1: Country cycles after removing the linear trend and quarterly means.

3 RE intercepts, common trend and quarterly dummies

2026-09-14 panel from 1995Q1: log real GDP per worker. 90 countries, 7975 observations, 1995-Q1–2026-Q2. Fitted countries: 90. Observations: 7975. Log-likelihood: 12732.16. Ergodic mean of the cycle $E[z] = -0.0437$ (0.0442). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -5.4306$ (0.0822), $\omega = 0.7551$ (0.0255). Posterior-mean a_i : mean -5.362, min -7.603, max -3.267. Common slope $g = 0.0166$ (0.0013). Seasonal dummies (Q1 omitted): $d_{Q2} = 0.0180$ (0.0008), $d_{Q3} = 0.0126$ (0.0009), $d_{Q4} = 0.0422$ (0.0008).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.2568 (0.0834)	0.0000 —	0.0408 (0.0899)
σ	0.1582 (0.0045)	0.0645 (0.0012)	0.0260 (0.0004)
ρ	0.6745 (0.0356)	0.9748 (0.0043)	0.9898 (0.0004)
Π_0	0.9814 (0.0061)	0.0121 (0.0051)	0.0065 (0.0049)
Π_1	0.0032 (0.0011)	0.9844 (0.0028)	0.0125 (0.0026)
Π_2	0.0018 (0.0010)	0.0114 (0.0022)	0.9868 (0.0020)
π	0.1168 (0.0260)	0.4243 (0.0382)	0.4589 (0.0391)

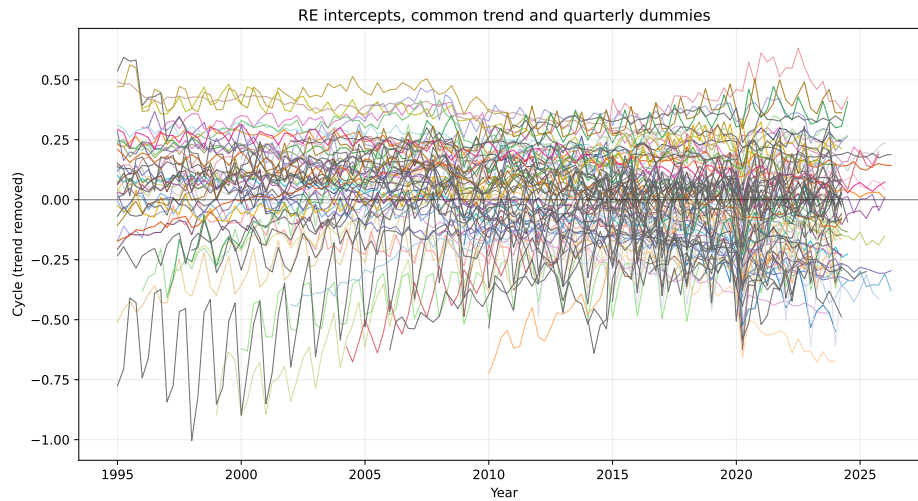


Figure 2: Country cycles after removing the linear trend and quarterly means.